

Project 2: Modeling Home Prices with the King County Housing Data

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Introduction

The King County Housing data set is a set of observations for thousands of houses in King County. Variables include zip code, square footage, number of bedrooms and bathrooms, among others. Modeling the relationship between home price and these other variables can give insight into the features of a home that add the most value, and provide a model that can make predictions of the price of homes in King County.

Methods

Data Collection

The data for this project was obtained from:

<https://lasanthi-asu.github.io/STT3851ClassRepo/Rmarkdown/Data/housedata.csv>

on May 5th, 2025 using the R programming language. From the available information, it's not clear when or how this data was originally collected, only that the data was collected by the King County Department of Assessments. The data was then split into a training and test set. The final model was trained using only the training set.

Exploratory Analysis

Exploratory Analysis was performed by examining the correlation table between the variables.

Statistical Modeling

Statistical models were made using linear regression, forward and backward subset selection, and ridge regression methods. The best was chosen based on test set performance. The final model is a linear model with most of the variables, an interaction term, and a quadratic term.

Reproducibility

All analyses performed in the project can be reproduced by running the `Project2Predictions.Rmd` file with RStudio. The R packages `DT` by Xie, Cheng, and Tan (2024), `boot` by Canty and Ripley (2023), `glmnet` by Friedman et al. (2023), `leaps` by Lumley (2022), `ggplot2` by Wickham et al. (2024), and `corrplot` by Wei and Simko (2021) will need to be installed.

Results

The final model consists of the following variables: `date`, `date^2`, `bedrooms`, `bathrooms`, `bedroomsxbathrooms`, `sqft_living`, `sqft_lot`, `floors`, `waterfront`, `view`, `condition`, `grade`, `sqft_above`, `yr_built`,

yr_renovated, zipcode, lat, and long. The variables grade, condition, and zipcode all have several levels.

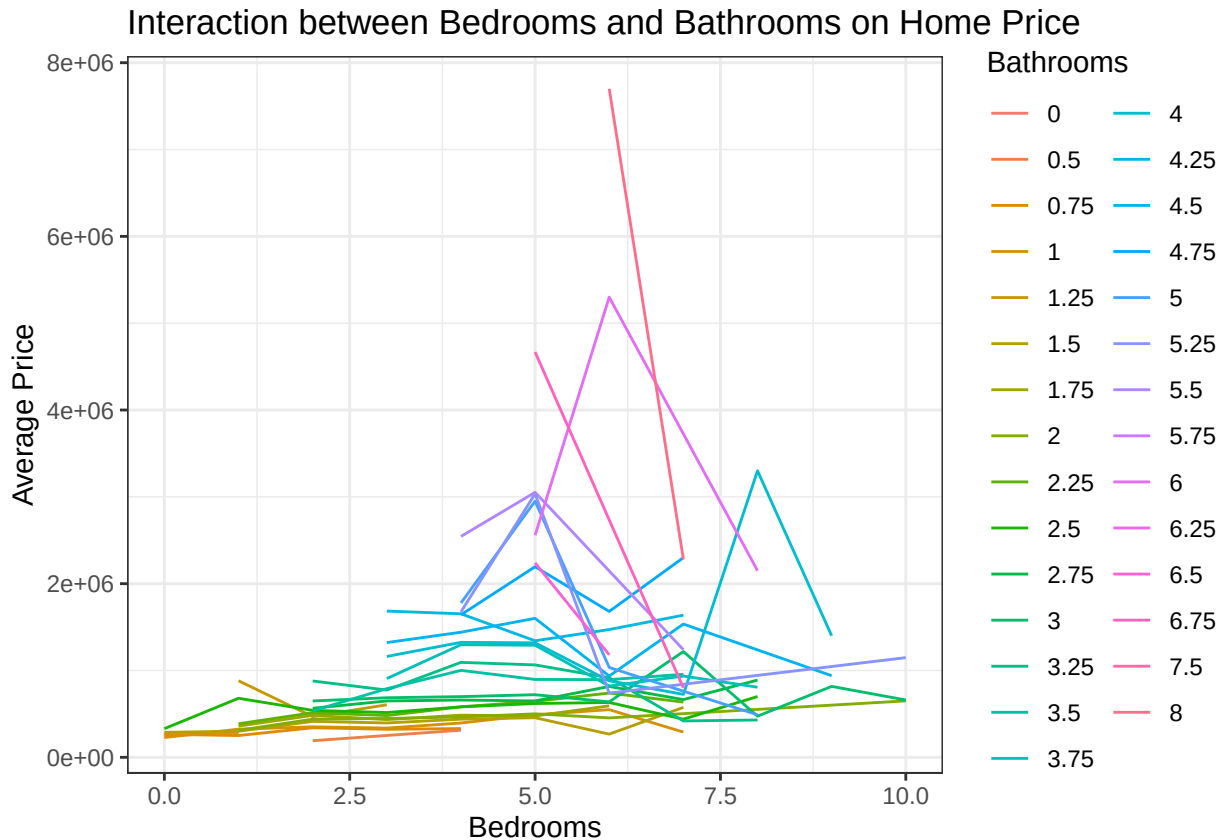


Figure 1: Interaction between bedrooms and bathrooms vs. home price. The non parallel lines suggest an interaction term between bedrooms and bathrooms will improve the model fit.

The interaction term was added based on the evidence from Figure 1. The quadratic term was added based on the evidence from Figure 2. A problem with the normal linear model is shown in Figure 3. The residual plot shows clear fanning out which indicates heteroscedasticity, meaning the error rate is not constant across the predictions.

To remedy this problem, I adjusted the model to predict the log of home prices instead of home prices alone. This change fixed the heteroscedasticity issue and produces the following residual plot in Figure 4 showing no issues.

The coefficients of the model are as follows:

```
##
## Call:
## lm(formula = price ~ . - sqft_lot15 + bedrooms * bathrooms +
##     poly(date, 2), data = housedata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1562051  -59679    2692   55806  3462814
##
## Coefficients: (2 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -4.235e+07  6.384e+06  -6.634 3.36e-11 ***
```

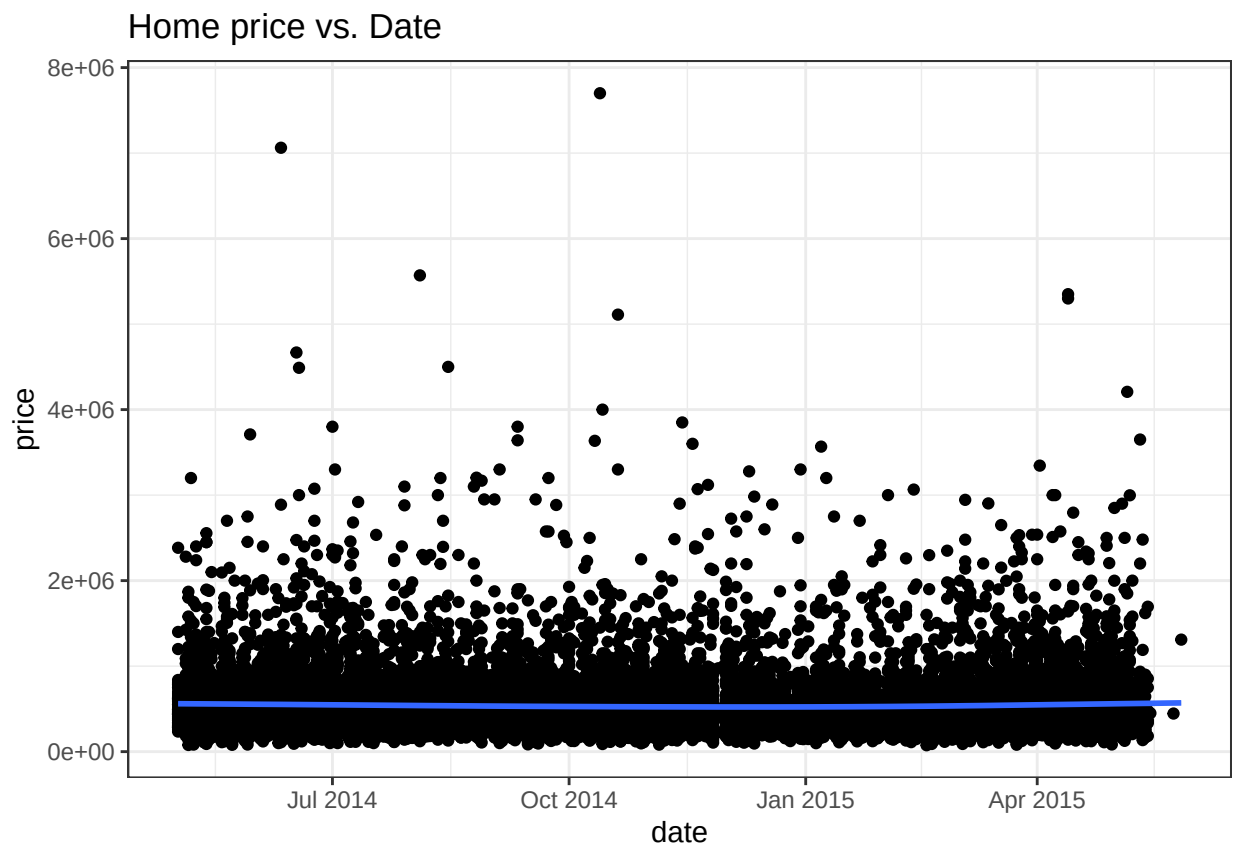


Figure 2: Home price predicted by date. Slight bowing of the line indicates the date variable is not perfectly linear and adding a quadratic term to it will increase model fit.

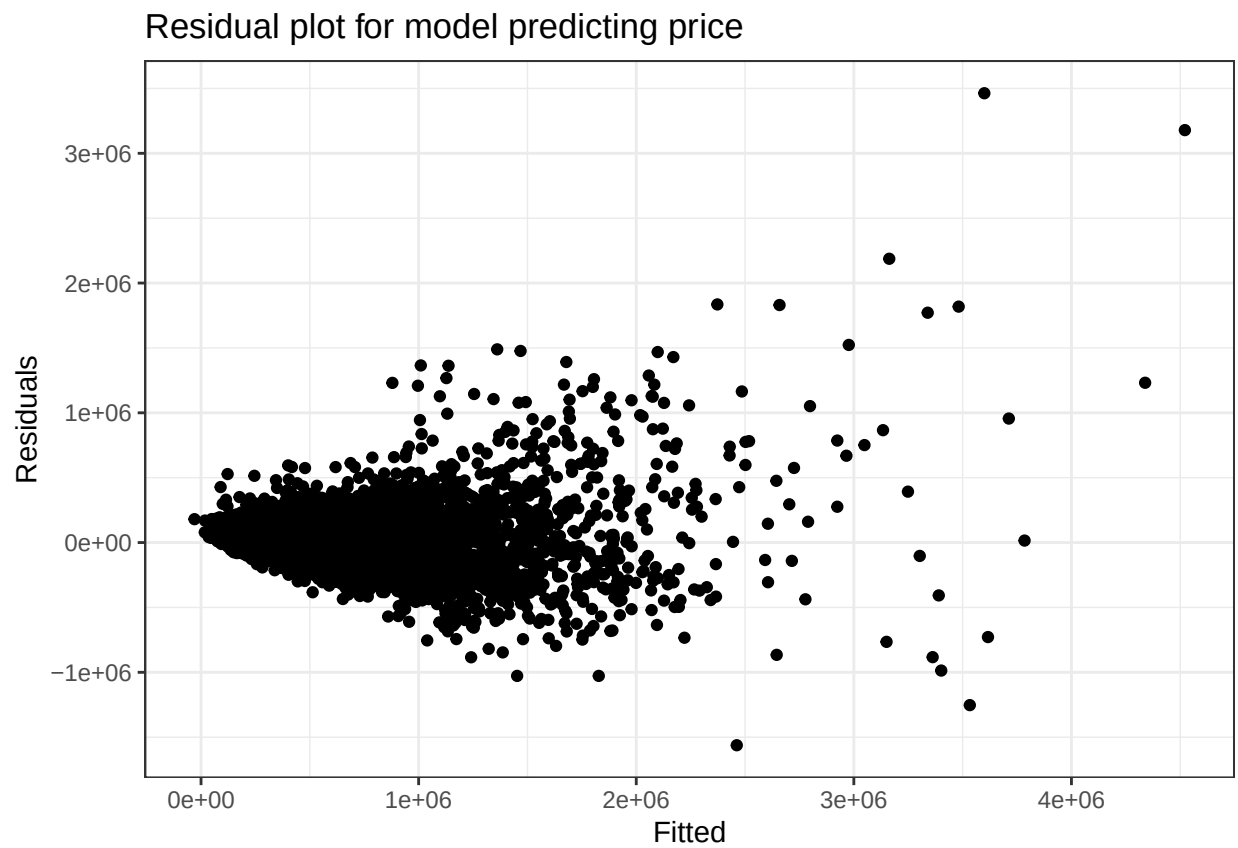


Figure 3: Residual plot for the linear model predicting home prices. The residual plot shows fanning which indicates nonconstant error across the predictions.

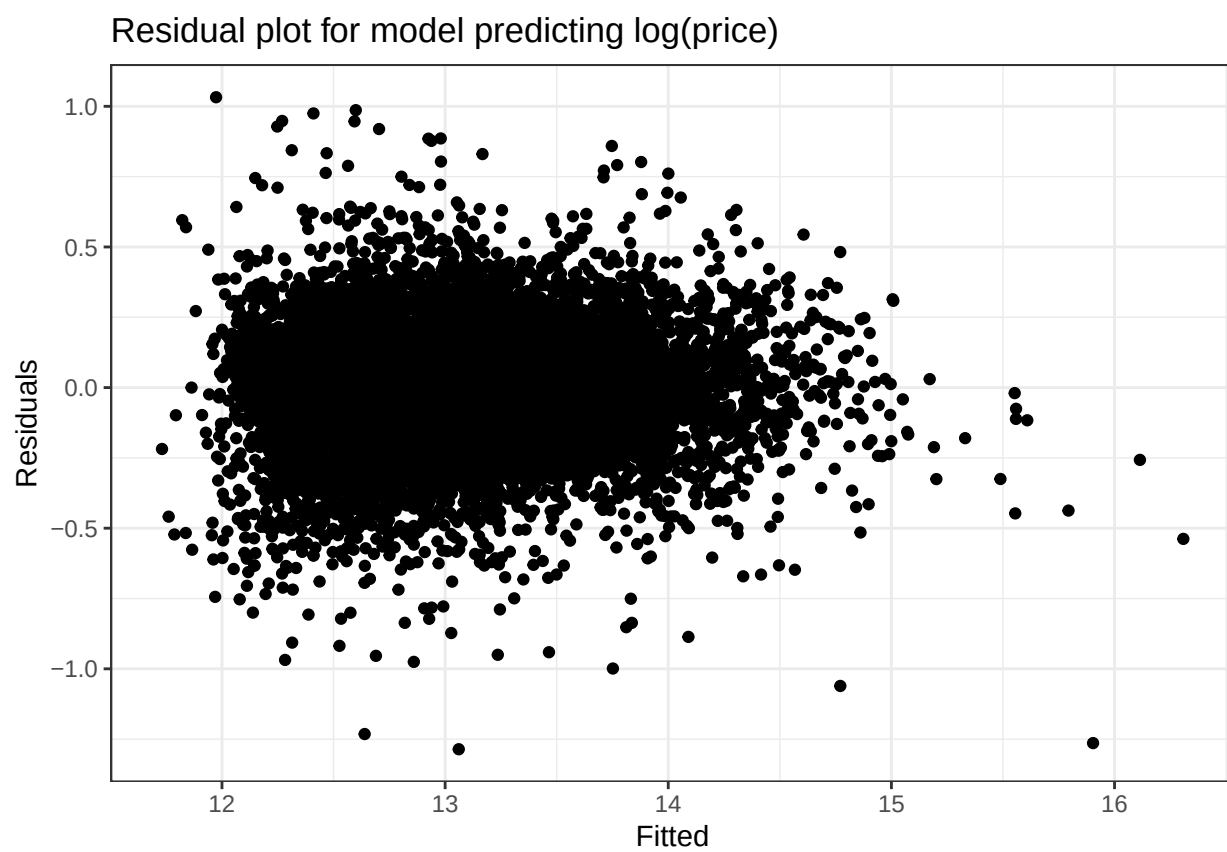


Figure 4: Residual plot for the linear model predicting the log of home prices. The fanning issue is resolved and the plot shows no clear issues.

## date	1.147e+02	1.013e+01	11.321	< 2e-16	***
## bedrooms	-3.083e+04	3.349e+03	-9.204	< 2e-16	***
## bathrooms	-4.260e+03	5.665e+03	-0.752	0.452112	
## sqft_living	1.159e+02	3.705e+00	31.282	< 2e-16	***
## sqft_lot	2.089e-01	3.036e-02	6.882	6.12e-12	***
## floors	-2.730e+04	3.355e+03	-8.138	4.31e-16	***
## waterfrontYes	6.658e+05	1.461e+04	45.570	< 2e-16	***
## view	4.938e+04	1.823e+03	27.085	< 2e-16	***
## conditionfair	9.979e+04	3.406e+04	2.930	0.003392	**
## conditionaverage	1.062e+05	3.165e+04	3.355	0.000797	***
## conditiongood	1.343e+05	3.167e+04	4.240	2.25e-05	***
## conditionvery good	1.856e+05	3.187e+04	5.825	5.83e-09	***
## grade10	1.647e+05	1.549e+05	1.063	0.287780	
## grade11	3.596e+05	1.551e+05	2.318	0.020472	*
## grade12	7.933e+05	1.560e+05	5.087	3.67e-07	***
## grade13	1.693e+06	1.612e+05	10.502	< 2e-16	***
## grade3	1.239e+04	1.771e+05	0.070	0.944215	
## grade4	-6.478e+04	1.585e+05	-0.409	0.682814	
## grade5	-6.391e+04	1.546e+05	-0.413	0.679411	
## grade6	-6.951e+04	1.546e+05	-0.450	0.652915	
## grade7	-6.174e+04	1.547e+05	-0.399	0.689745	
## grade8	-3.924e+04	1.547e+05	-0.254	0.799786	
## grade9	3.605e+04	1.548e+05	0.233	0.815836	
## sqft_above	5.065e+01	3.858e+00	13.130	< 2e-16	***
## sqft_basement	NA	NA	NA	NA	
## yr_built	-3.286e+00	2.791e-01	-11.773	< 2e-16	***
## yr_renovated	2.609e+00	2.840e-01	9.188	< 2e-16	***
## zipcode98002	1.856e+04	1.512e+04	1.228	0.219623	
## zipcode98003	-1.142e+04	1.338e+04	-0.853	0.393468	
## zipcode98004	7.158e+05	2.449e+04	29.232	< 2e-16	***
## zipcode98005	2.554e+05	2.619e+04	9.750	< 2e-16	***
## zipcode98006	2.206e+05	2.134e+04	10.339	< 2e-16	***
## zipcode98007	2.125e+05	2.683e+04	7.919	2.53e-15	***
## zipcode98008	2.396e+05	2.561e+04	9.355	< 2e-16	***
## zipcode98010	1.113e+05	2.274e+04	4.893	1.00e-06	***
## zipcode98011	5.975e+04	3.349e+04	1.784	0.074456	.
## zipcode98014	1.024e+05	3.700e+04	2.766	0.005673	**
## zipcode98019	7.350e+04	3.604e+04	2.039	0.041431	*
## zipcode98022	7.243e+04	2.005e+04	3.612	0.000305	***
## zipcode98023	-5.347e+04	1.236e+04	-4.325	1.54e-05	***
## zipcode98024	1.808e+05	3.175e+04	5.694	1.26e-08	***
## zipcode98027	1.723e+05	2.197e+04	7.840	4.77e-15	***
## zipcode98028	3.714e+04	3.248e+04	1.143	0.252904	
## zipcode98029	2.364e+05	2.501e+04	9.453	< 2e-16	***
## zipcode98030	1.336e+04	1.460e+04	0.915	0.360428	
## zipcode98031	1.598e+04	1.532e+04	1.043	0.297180	
## zipcode98032	-1.119e+04	1.787e+04	-0.626	0.531160	
## zipcode98033	2.955e+05	2.781e+04	10.625	< 2e-16	***
## zipcode98034	1.278e+05	2.982e+04	4.284	1.85e-05	***
## zipcode98038	7.979e+04	1.650e+04	4.836	1.34e-06	***
## zipcode98039	1.100e+06	3.237e+04	33.979	< 2e-16	***
## zipcode98040	4.645e+05	2.189e+04	21.219	< 2e-16	***
## zipcode98042	2.868e+04	1.413e+04	2.029	0.042476	*
## zipcode98045	1.840e+05	3.057e+04	6.021	1.77e-09	***

```

## zipcode98052      1.972e+05  2.834e+04   6.960 3.53e-12 ***
## zipcode98053      1.816e+05  3.029e+04   5.996 2.06e-09 ***
## zipcode98055      2.199e+04  1.713e+04   1.284 0.199257
## zipcode98056      6.374e+04  1.862e+04   3.423 0.000621 ***
## zipcode98058      3.407e+04  1.616e+04   2.109 0.034967 *
## zipcode98059      6.979e+04  1.832e+04   3.810 0.000139 ***
## zipcode98065      1.365e+05  2.804e+04   4.868 1.14e-06 ***
## zipcode98070     -7.940e+04  2.149e+04  -3.696 0.000220 ***
## zipcode98072      9.112e+04  3.312e+04   2.751 0.005949 **
## zipcode98074      1.605e+05  2.677e+04   5.995 2.07e-09 ***
## zipcode98075      1.603e+05  2.567e+04   6.246 4.32e-10 ***
## zipcode98077      6.052e+04  3.459e+04   1.750 0.080198 .
## zipcode98092     -2.155e+03  1.322e+04  -0.163 0.870508
## zipcode98102      4.382e+05  2.927e+04  14.968 < 2e-16 ***
## zipcode98103      2.376e+05  2.703e+04   8.793 < 2e-16 ***
## zipcode98105      3.967e+05  2.767e+04  14.341 < 2e-16 ***
## zipcode98106      5.269e+04  2.001e+04   2.633 0.008476 **
## zipcode98107      2.449e+05  2.773e+04   8.831 < 2e-16 ***
## zipcode98108      5.938e+04  2.231e+04   2.661 0.007794 **
## zipcode98109      4.123e+05  2.854e+04  14.448 < 2e-16 ***
## zipcode98112      5.360e+05  2.537e+04  21.124 < 2e-16 ***
## zipcode98115      2.447e+05  2.737e+04   8.941 < 2e-16 ***
## zipcode98116      2.047e+05  2.227e+04   9.195 < 2e-16 ***
## zipcode98117      2.144e+05  2.776e+04   7.723 1.20e-14 ***
## zipcode98118      1.097e+05  1.943e+04   5.646 1.67e-08 ***
## zipcode98119      4.029e+05  2.695e+04  14.953 < 2e-16 ***
## zipcode98122      2.698e+05  2.416e+04  11.169 < 2e-16 ***
## zipcode98125      1.081e+05  2.961e+04   3.651 0.000262 ***
## zipcode98126      1.173e+05  2.063e+04   5.687 1.31e-08 ***
## zipcode98133      5.028e+04  3.053e+04   1.647 0.099652 .
## zipcode98136      1.734e+05  2.095e+04   8.276 < 2e-16 ***
## zipcode98144      2.171e+05  2.239e+04   9.695 < 2e-16 ***
## zipcode98146      2.856e+04  1.871e+04   1.527 0.126876
## zipcode98148      3.385e+04  2.628e+04   1.288 0.197738
## zipcode98155      4.211e+04  3.183e+04   1.323 0.185968
## zipcode98166      4.821e+03  1.721e+04   0.280 0.779345
## zipcode98168      6.937e+03  1.821e+04   0.381 0.703236
## zipcode98177      1.067e+05  3.182e+04   3.353 0.000801 ***
## zipcode98178     -4.591e+03  1.856e+04  -0.247 0.804648
## zipcode98188     -5.779e+03  1.893e+04  -0.305 0.760121
## zipcode98198     -3.223e+04  1.444e+04  -2.232 0.025640 *
## zipcode98199      2.700e+05  2.633e+04  10.257 < 2e-16 ***
## lat               2.242e+05  6.614e+04   3.390 0.000700 ***
## long             -2.440e+05  4.704e+04  -5.186 2.17e-07 ***
## sqft_living15     1.554e+01  3.019e+00   5.148 2.66e-07 ***
## poly(date, 2)1      NA          NA          NA          NA
## poly(date, 2)2     1.181e+06  1.504e+05   7.853 4.28e-15 ***
## bedrooms:bathrooms 7.722e+03  1.350e+03   5.719 1.09e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 149800 on 17283 degrees of freedom
## Multiple R-squared:  0.8368, Adjusted R-squared:  0.8358
## F-statistic: 886 on 100 and 17283 DF, p-value: < 2.2e-16

```

$$\log(\hat{HomePrice}) = -42350000 + 114.7date + \dots + 7722bedrooms : bathrooms$$

The following ANOVA analysis of the model shows every variable has a statistically significant relationship with $\log(\text{price})$.

```
## Analysis of Variance Table
##
## Response: log(price)
##
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
date	1	0.13	0.13	4.1040	0.04280 *
bedrooms	1	582.85	582.85	18169.1084	< 2.2e-16 ***
bathrooms	1	892.15	892.15	27811.0035	< 2.2e-16 ***
sqft_living	1	896.46	896.46	27945.1370	< 2.2e-16 ***
sqft_lot	1	2.56	2.56	79.8214	< 2.2e-16 ***
floors	1	10.38	10.38	323.6854	< 2.2e-16 ***
waterfront	1	47.37	47.37	1476.7601	< 2.2e-16 ***
view	1	75.72	75.72	2360.3183	< 2.2e-16 ***
condition	4	52.21	13.05	406.9032	< 2.2e-16 ***
grade	11	303.21	27.56	859.2751	< 2.2e-16 ***
sqft_above	1	20.97	20.97	653.7256	< 2.2e-16 ***
yr_built	1	263.36	263.36	8209.7191	< 2.2e-16 ***
yr_renovated	1	0.18	0.18	5.6401	0.01757 *
zipcode	69	1089.10	15.78	492.0371	< 2.2e-16 ***
lat	1	1.56	1.56	48.5361	3.360e-12 ***
long	1	1.81	1.81	56.4265	6.118e-14 ***
sqft_living15	1	18.18	18.18	566.8741	< 2.2e-16 ***
poly(date, 2)	1	4.21	4.21	131.1259	< 2.2e-16 ***
bedrooms:bathrooms	1	5.65	5.65	176.0198	< 2.2e-16 ***
Residuals	17283	554.42	0.03		

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions

```
## Predicted Lower_PI Upper_PI
## 1 302890.6 182766.3 501967.4
## 2 788795.9 554723.8 1121637.5
## 3 227694.6 160066.2 323896.1
## 4 459844.4 306037.1 690951.8
## 5 444161.5 295711.8 667134.2
## 6 466908.5 328462.6 663708.8
```

This prediction table shows the model's first 6 predicted home price values on the test data set. The results of the model predictions were exponentiated to convert from log back to normal prices. An upper and lower prediction interval is also given.

References

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